TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Carpet Testing

Thickness, compression and recovery, dynamic loading and tuft bind — how a pile floor covering is judged for the years of footsteps it must survive.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 10 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the key performance properties of a tufted carpet: total thickness and pile height under standard pressures, thickness loss under dynamic loading and its recovery after rest, tuft-bind force (in the principle of ISO 4919 — determination of tuft withdrawal force), and tuft density, and to interpret the results for carpet grading and end-use selection.

1.1 The carpet as a structure

A tufted carpet is a layered composite, and every test in this sheet addresses one layer or one interface (see Figure 1). The wearing surface is the **pile**: yarns — cut to form cut pile, or left as loops in loop pile — standing in a **primary backing** (usually woven polypropylene) through which they were tufted. A **latex coat** locks the tufts into the backing from behind; its quality largely decides tuft bind. A **secondary backing** (jute or synthetic) is laminated on for dimensional stability and handling. Underfoot comfort, insulation, appearance retention and wear all depend on the pile — its height, density, fibre and resilience — while the backings decide whether the carpet lies flat, keeps its tufts and survives installation.

Because a carpet is compressible by design, even its thickness is a defined quantity, not a single obvious number: thickness is measured under a stated pressure applied for a stated time, and pile height is obtained as the difference between total thickness and the thickness of the backing measured under higher pressure that compresses the pile flat. Specifications that omit the pressure are meaningless, and the first discipline of carpet testing is to quote every thickness with its pressure.

1.2 Compression, dynamic loading and recovery

Walking subjects a carpet to thousands of short, heavy local loads — a heel strike is a severe impact on a few square centimetres. Two laboratory responses are measured. In a **static compression and recovery** check, thickness is followed as pressure is increased and released: the compression curve shows how readily the pile yields (comfort), and the recovery on unloading shows how much of the

deformation is elastic. In **dynamic loading**, a weighted striker with standard feet impacts the specimen repeatedly — 1,000 impacts in the worked example below — simulating concentrated traffic. The thickness loss after loading is the headline durability figure:

$$\text{Thickness loss under loading (\%)} = (\text{initial thickness} - \text{thickness after loading}) / \text{initial thickness} \times 100$$

Recovery matters as much as loss. Pile fibres bend under load and recover with time; a carpet measured immediately after loading looks worse than it will after a night's rest. The sheet therefore measures again after a rest period and separates the loss into a recoverable part and a permanent part:

$$\text{Permanent thickness loss (\%)} = (\text{initial thickness} - \text{thickness after rest}) / \text{initial thickness} \times 100$$

Resilient fibres (wool pre-eminently, then quality nylon) and dense, well-supported pile show small permanent loss: the tufts are crowded enough to hold each other upright and the fibre itself springs back. Sparse pile of a soft fibre flattens permanently — the familiar crushed traffic lane — even if it has not yet worn away. Appearance retention grading in commercial practice formalises exactly this behaviour over longer test programmes.

1.3 Tuft bind and tuft density

Tuft bind is the force needed to pull a single tuft (or loop) straight out of the backing. It tests the latex anchor: a weak bind sheds tufts in vacuuming, snags and pulls in use, and ruins loop-pile carpet by laddering when one loop is caught. In the ISO 4919 principle, one leg of a cut tuft (or one loop) is clamped and withdrawn vertically at a standard rate on a tensile tester, and the peak force is recorded for a series of tufts spread across the specimen. Low individual readings matter as well as the mean — one badly anchored row is where failure starts.

Tuft density — tufts per unit area — is counted directly: tufts per inch along the row (the machine gauge direction) multiplied by rows (stitches) per inch along the length. Together with pile height and yarn count it sets the pile mass per unit area, the single best construction predictor of wear life: more fibre in the wear layer, more service before the backing grins through. Density also explains compression results — a dense pile resists flattening because neighbouring tufts support each other — so the four determinations of this sheet are read as one story, not four unrelated numbers. All specimens are conditioned and tested in the standard atmosphere (65% RH, 20 °C); wool-rich piles in particular change resilience markedly with regain.

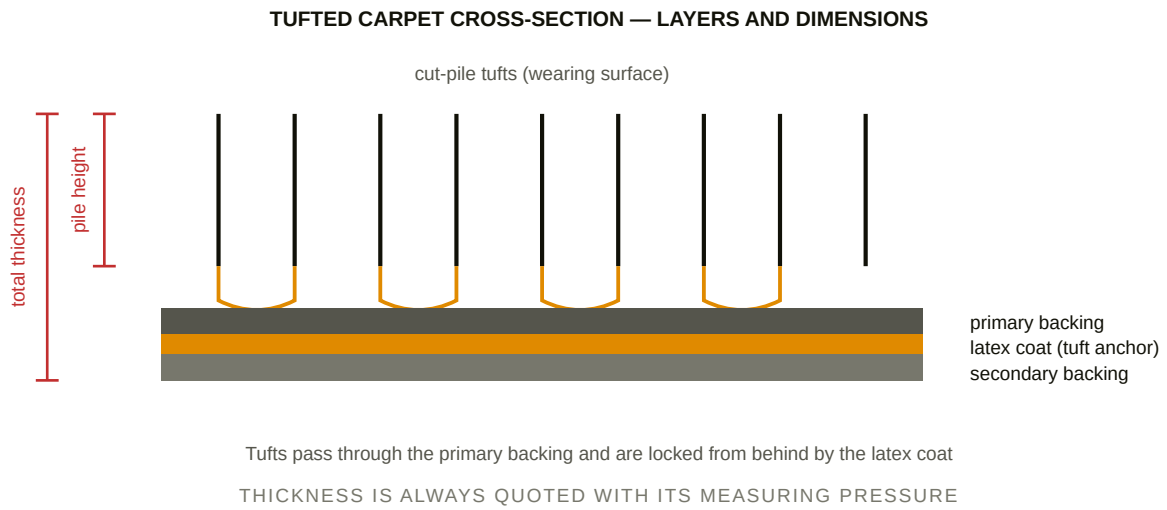


Figure 1 — Cross-section of a tufted carpet: cut-pile tufts standing in the primary backing, the U-shaped tuft legs anchored by the latex coat, and the secondary backing beneath. Pile height and total thickness are dimensioned separately — their difference is the backing thickness.

2. Apparatus and procedure

- **Apparatus:** carpet thickness gauge with interchangeable presser feet and known pressures; dynamic loading machine (weighted striker with standard feet and impact counter); tensile tester with tuft-withdrawal clamp (ISO 4919 principle); counting glass and ruler for tuft density; sharp blade for exposing tufts; conditioning facilities (65% RH, 20 °C).
- **Conditioning and marking:** condition specimens in the standard atmosphere (65% RH, 20 °C). Mark fixed measuring points on each specimen so that every thickness in the sequence — initial, loaded, recovered — is read at the same places; pile is not uniform enough for wandering measurement points.
- **Thickness:** measure total thickness at the marked points under the standard low pressure and record the mean as the initial thickness. (Where pile height is required, re-measure under the higher pressure that flattens the pile and subtract — the standard difference method.)
- **Dynamic loading:** mount the specimen, apply 1,000 impacts of the standard striker, remove the specimen without brushing or steaming the pile, and re-measure thickness at the marked points in the impacted zone. Rest the specimen flat and unloaded for the standard recovery period (24 h in commercial practice; the period used must be stated) and measure a third time.
- **Tuft bind:** expose individual tufts at spread positions across a fresh specimen. For cut pile, clamp one leg of a tuft; for loop pile, hook one loop. Withdraw vertically at the standard rate and record the peak force. Test at least five tufts (more for a specification decision) and report each value and the mean, noting any tuft that slipped from a broken latex anchor at very low force.

- **Tuft density:** count tufts in a measured inch along a row and rows in a measured inch along the length, at several places away from edges; multiply the two means for tufts per square inch. Record gauge and stitch rate alongside.

3. Worked calculation

Dynamic loading. Mean thicknesses at the marked points: initial 9.6 mm; immediately after 1,000 impacts 8.1 mm; after the rest period 8.7 mm.

$$\text{Thickness loss after loading} = (9.6 - 8.1) / 9.6 \times 100 = 1.5 / 9.6 \times 100 = 15.6\%$$

$$\text{Recovered thickness} = 8.7 - 8.1 = 0.6 \text{ mm of the } 1.5 \text{ mm lost} \quad (40\% \text{ of the loss recovered})$$

$$\text{Permanent thickness loss} = (9.6 - 8.7) / 9.6 \times 100 = 0.9 / 9.6 \times 100 = 9.4\%$$

The distinction between the two loss figures is the point of the exercise. Judged immediately after loading, the carpet has lost 15.6% of its thickness — an alarming figure if reported alone. After rest, 0.6 mm has sprung back and the permanent loss is 9.4%: the pile fibres were bent, not broken or displaced, and recovered two-fifths of the deformation unaided. A permanent loss under about 10% after this loading is a respectable domestic result; a contract carpet for heavy traffic is expected to do better, and a sparse, low-density pile would typically show a permanent loss well into double figures with little recovery.

Tuft bind. Five withdrawal peaks: 28, 31, 30, 27 and 29 N.

$$\text{Mean tuft bind} = (28 + 31 + 30 + 27 + 29) / 5 = 145 / 5 = 29 \text{ N}$$

The mean of 29 N is firm anchoring for a domestic tufted carpet, and — as important — the lowest individual value, 27 N, is close to the mean: there is no weak row or starved latex patch in the area sampled. A mean carried by four high values with one value near zero would fail the carpet in practice even at the same average.

Tuft density. Counted 8 tufts per inch along the row and 9 rows per inch.

$$\text{Tuft density} = 8 \times 9 = 72 \text{ tufts/in}^2$$

> **RESULT AS ONE STORY**
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> Initial thickness 9.6 mm; loss after 1,000 impacts 15.6%, recovering to a permanent loss of 9.4% after rest; mean tuft bind 29 N with a tight 27–31 N spread; density 72 tufts/in². A moderately dense pile of resilient behaviour, firmly anchored — fit for general domestic use; heavy contract use would ask for higher density and lower permanent loss.

4. Precautions

- Quote every thickness with its presser-foot pressure and dwell time, and always re-measure at the marked points. A carpet has no single thickness independent of how it is squeezed.
- Do not brush, vacuum, steam or finger-lift the pile between loading and the after-loading measurement — grooming is recovery assistance the traffic in service will not provide on schedule, and it falsifies the loss figure.
- State the rest period before the recovery measurement and keep it the same for every specimen compared. Recovery is time-dependent; a 1-hour rest and a 24-hour rest give different "permanent" losses.
- In tuft bind, withdraw truly vertically at the standard rate and clamp a single tuft leg (or single loop). An angled pull levers the tuft against the backing and reads high; clamping two legs reads roughly double and is a different test.
- Count density away from cut edges and pattern boundaries, at several places. Edge rows are distorted by cutting, and a single count in a carpet is an anecdote, not a density.
- Condition wool-containing carpets fully. A dry wool pile tests harsher and less resilient than the same pile at 65% RH, and the comparison between laboratories collapses.

Practice (viva preparation)

1. Name the four layers of a tufted carpet and the function of the latex coat.
2. A carpet is 9.6 mm thick initially and 8.1 mm after 1,000 dynamic-loading impacts. Find the thickness loss percentage.
3. After rest the same carpet recovers to 8.7 mm. Find the permanent thickness loss and explain why it differs from the loss in Question 2.
4. Tuft-bind readings are 28, 31, 30, 27 and 29 N. Find the mean and say why the lowest value matters.
5. Counts give 8 tufts per inch along the row and 9 rows per inch. Find the tuft density and state what density predicts.
6. Why must a thickness value always be quoted with its measuring pressure?

Answer key

1. Pile (the wearing surface of tufted yarn), primary backing (the woven sheet the tufts are inserted through), latex coat, and secondary backing (stability and handling). The latex coat is the anchor: it locks each tuft leg into the primary backing and so determines tuft bind, shedding and resistance to loop pull-out.
2. Loss = (9.6 − 8.1) / 9.6 × 100 = 1.5 / 9.6 × 100 = 15.6%.

3. Permanent loss = (9.6 − 8.7) / 9.6 × 100 = 0.9 / 9.6 × 100 = 9.4%. The difference is elastic recovery: under impact the pile fibres bend and the structure compacts; given unloaded rest, resilient fibres straighten and 0.6 mm of the 1.5 mm loss returns. Only the unrecovered 0.9 mm — fibre set, displacement and structural settling — is permanent. Reporting only the immediate figure overstates the damage done by traffic.
4. Mean = (28 + 31 + 30 + 27 + 29) / 5 = 145 / 5 = 29 N. The lowest value (27 N) matters because failure starts at the weakest anchor — a single starved or broken latex patch sheds tufts and starts laddering regardless of a healthy mean — and here the tight spread shows uniformly sound anchoring.
5. Density = 8 × 9 = 72 tufts/in². With pile height and yarn count, density sets pile mass per unit area — the depth of wearing material — and so predicts wear life and resistance to flattening: denser piles support each tuft against its neighbours and keep their appearance longer under traffic.
6. Carpet pile is a compressible fibre bed: thickness falls continuously as pressure rises, and different pressures flatten different fractions of the pile. Two gauges at different pressures therefore measure different "thicknesses" of the same specimen, and comparison — between specimens, or against a specification — is only possible at an agreed pressure and dwell time.

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